

Lecture 3: Random Matrices (Preliminaries)

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- What is a **random matrix**? Basic objects: **eigenvalues**, **singular values**.
- Canonical families: **Wigner** and **Wishart**.
- Foundational tools: **Min–Max**, **Norms**.
- Asymptotics: **Semicircle Law** (and preview of **Marchenko–Pastur**).

Why Random Matrices Matter

A matrix as a transformation

A matrix acts like a machine that takes a vector and **changes its shape**: it can stretch, squeeze, or rotate space.

Why randomness helps

- ▶ They help us understand how **lengths and angles** behave on average.
- ▶ They appear naturally when studying **noise, data, and stability** of algorithms.

Basic tools: Eigenvalues and Eigenvectors

We start with the definition in a real vector space (the same extends to the complex case).

Definition 1

For a square matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, $\lambda \in \mathbb{R}$ is an eigenvalue of $\mathbf{A}$ and a $\mathbf{v} \neq 0$ is an eigenvector of $\mathbf{A}$ if the following holds

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

Equivalently, $(\mathbf{A} - \lambda I)\mathbf{v} = 0$ or $\det(\mathbf{A} - \lambda I) = 0$.

Eigenvalues and Eigenvectors – tiny example

Diagonal matrix

Consider

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

$\mathbf{A}e_1 = 2e_1 \Rightarrow \text{eigenpair } (\lambda = 2, e_1); \mathbf{A}e_2 = 1e_2 \Rightarrow (\lambda = 1, e_2).$

Spectral Theorem

Statement

If $\mathbf{A} \in \mathbb{R}^{d \times d}$ is **symmetric**, then:

$$\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top,$$

where $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_d]$ is an **orthogonal matrix** ($\mathbf{V}^\top \mathbf{V} = \mathbf{I}$) whose columns are the **eigenvectors** of $\mathbf{A}$, and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_d)$ contains the real eigenvalues of $\mathbf{A}$.

Geometric meaning

The matrix $\mathbf{A}$ acts by

$$\mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{v}_i,$$

so it stretches or compresses space along d **orthogonal directions**.

(If $\mathbf{A}$ is **diagonal**, then $\mathbf{V} = \mathbf{I}$ and $\mathbf{\Lambda} = \mathbf{A}$: the eigenvectors are the standard basis vectors)

Singular values – definition

Now consider a rectangular $m \times n$ matrix $\mathbf{A}$ with real entries.

Definition 2

The i -th singular value of $\mathbf{A}$ is

$$s_i(\mathbf{A}) = \sqrt{\lambda_i(\mathbf{A}\mathbf{A}^\top)} = \sqrt{\lambda_i(\mathbf{A}^\top\mathbf{A})}.$$

Singular values and eigenvalues

If $\mathbf{A}$ is a symmetric (square) matrix then the singular values of $\mathbf{A}$ are the absolute values of the eigenvalues λ_i of $\mathbf{A}$: $s_i(\mathbf{A}) = |\lambda_i(\mathbf{A})|$

Singular Value Decomposition (SVD)

Statement

For any (possibly rectangular) matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$,

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top,$$

where

- ▶ $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$ are **orthogonal matrices**,
- ▶ $\mathbf{\Sigma} = \text{diag}(s_1, \dots, s_r)$ with $s_1 \geq \dots \geq s_r > 0$ contains the **singular values** of $\mathbf{A}$.

Remark

- ▶ The columns of $\mathbf{V}$ (the **right singular vectors**) are eigenvectors of $\mathbf{A}^\top \mathbf{A}$.
- ▶ The columns of $\mathbf{U}$ (the **left singular vectors**) are eigenvectors of $\mathbf{A} \mathbf{A}^\top$.

Random Matrix – definition

Definition 3 (random matrix)

An $m \times n$ matrix $\mathbf{X} = [x_{ij}]$ is random if its entries x_{ij} are random variables.

Remark

Each draw of the entries gives a **new matrix** – hence **new eigenvalues** and **singular values**.

(see J. A. Tropp, *An introduction to matrix concentration inequalities*, for a thorough intro)

A simple example: a 2×2 random matrix

Gaussian example

Let each entry be drawn independently from a standard normal distribution:

$$x_{ij} \sim \mathcal{N}(0, 1), \quad \mathbf{X} = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}.$$

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One random realization

$$\mathbf{X} = \begin{pmatrix} 0.5 & -1.3 \\ 1.1 & 0.2 \end{pmatrix} \quad (\text{next time it will look different!})$$

- ▶ Random eigenvalues and singular values;
- ▶ Random geometric effects (stretching, rotation) at each draw.

Empirical Spectral Distribution (ESD)

Definition

For eigenvalues $\lambda_1(\mathbf{X}), \dots, \lambda_n(\mathbf{X})$,

$$\mu_{\mathbf{X}} = \frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i(\mathbf{X})}.$$

Empirical Spectral Distribution (ESD)

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- $\mu_{\mathbf{X}}$ is a **probability distribution** on $\mathbb{R}$. It places equal mass on each eigenvalue of $\mathbf{X}$:

$$\mu_{\mathbf{X}}([a, b]) = \frac{1}{n} \#\{i : \lambda_i(\mathbf{X}) \in [a, b]\}.$$

- For any function f ,

$$\int f(\lambda) d\mu_{\mathbf{X}}(\lambda) = \frac{1}{n} \sum_{i=1}^n f(\lambda_i(\mathbf{X})).$$

- In random matrix theory we study $\mu_{\mathbf{X}} \xrightarrow{n \rightarrow \infty}$ (limiting law, e.g. semicircle).

ESD – micro-example

2×2 diagonal

If $\mathbf{X} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$, eigenvalues are $\{2, -1\}$. Then

$$\mu_{\mathbf{X}} = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_2.$$

Remark

Repeated eigenvalues get repeated mass in the ESD.
(Yes—**multiplicity counts.**)

Example of random matrices: Wigner Matrices

Definition

A **Wigner matrix** is a random symmetric matrix

$$\mathbf{X} = (x_{ij})_{1 \leq i, j \leq n},$$

where the entries x_{ij} are **independent**, **zero-mean**, and **identically distributed** random variables. In this lecture we consider **real entries**, but complex versions also exist.

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Examples

- ▶ **Symmetric Bernoulli Ensemble:** $x_{ij} \in \{\pm 1\}$ with equal probability, and $x_{ij} = x_{ji}$.
- ▶ **Gaussian Orthogonal Ensemble (GOE):** $x_{ij} \sim \mathcal{N}(0, 1)$ for $i < j$, $x_{ii} \sim \mathcal{N}(0, 2)$, and $x_{ij} = x_{ji}$.

(as $n \rightarrow \infty$, the ESD of $\mathbf{X}$ is described by the **Wigner semicircle law**)

Example of Random Matrices: Wishart Matrices

Definition

Let

$$\mathbf{R} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix}$$

be an $m \times n$ random matrix whose **columns** are independent samples from an m -variate normal distribution $\mathcal{N}(0, I_m)$. The associated **Wishart matrix** is

$$\mathbf{C}_m = \frac{1}{m} \mathbf{R} \mathbf{R}^\top \in \mathbb{R}^{m \times m}.$$

Spectral behavior

As $m, n \rightarrow \infty$, $m/n \rightarrow \gamma \in (0, \infty)$, the ESD of $\mathbf{C}_m$ converges to the **Marchenko–Pastur law**.

Questions We Will Address (Today and Next Class)

Our goal is to understand some of the **fundamental properties of random matrices**, and how they connect to core ideas in data science.

- What is the **expected behavior of the singular values** of random matrices, in particular the **smallest** and **largest** ones?
- For **symmetric square matrices** (such as Wigner or Wishart ensembles), what can we say about the **expected extreme eigenvalues** and their typical fluctuations?
- How are these spectral properties **used in fundamental algorithms** such as random projection methods and **principal component analysis (PCA)**?

Eigenvalues and notation

Recall

For a square matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, an eigenpair $(\lambda, \mathbf{v})$ satisfies

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Sorted order

We order the eigenvalues of $\mathbf{A}$ as

$$\lambda_1(\mathbf{A}) \geq \lambda_2(\mathbf{A}) \geq \dots \geq \lambda_d(\mathbf{A}),$$

so that we can refer to the **largest** and **smallest** eigenvalues as $\lambda_1(\mathbf{A})$ and $\lambda_d(\mathbf{A})$, respectively.

Min–Max Principle for Eigenvalues – Setup

Hermitian matrices

A matrix $\mathbf{A} \in \mathbb{C}^{n \times n}$ is **Hermitian** if

$$\mathbf{A} = \overline{\mathbf{A}^\top}.$$

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Subspaces and dimension

A **subspace** $V \subseteq \mathbb{R}^n$ is any linear space spanned by some vectors of $\mathbb{R}^n$. The **dimension** $\dim(V)$ is the number of vectors in any basis of V .

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Theorem 1 (Courant–Fischer Min–Max Principle)

For each $i = 1, \dots, n$,

$$\lambda_i(\mathbf{A}) = \min_{\dim(V)=n-i+1} \max_{\substack{\mathbf{x} \in V \\ \|\mathbf{x}\|_2=1}} \mathbf{x}^\top \mathbf{A} \mathbf{x} = \max_{\dim(V)=i} \min_{\substack{\mathbf{x} \in V \\ \|\mathbf{x}\|_2=1}} \mathbf{x}^\top \mathbf{A} \mathbf{x}.$$

Min-Max Principle – Extremal Values

Extremal cases

- ▶ The Min-Max principle yields the following expression for the **largest** and **smallest** eigenvalues:

$$\lambda_1(\mathbf{A}) = \max_{\|\mathbf{x}\|_2=1} \mathbf{x}^\top \mathbf{A} \mathbf{x}, \quad \lambda_n(\mathbf{A}) = \min_{\|\mathbf{x}\|_2=1} \mathbf{x}^\top \mathbf{A} \mathbf{x}.$$

Min-Max for Singular Values

Theorem 2 (Min-Max Theorem for Singular Values)

Let $\mathbf{A}$ be an $m \times n$ Hermitian matrix.

We denote by $s_i(\mathbf{A})$ the i -th singular value of $\mathbf{A}$, which is equal to

$$s_i(\mathbf{A}) = \sqrt{\lambda_i(\mathbf{A}\mathbf{A}^\top)} = \sqrt{\lambda_i(\mathbf{A}^\top\mathbf{A})}.$$

Let V be a subspace of $\mathbb{R}^n$. Then

$$s_i(\mathbf{A}) = \min_{\dim(V)=n-i+1} \max_{\substack{\mathbf{x} \in V \\ \|\mathbf{x}\|_2=1}} \|\mathbf{A}\mathbf{x}\|_2 = \max_{\dim(V)=i} \min_{\substack{\mathbf{x} \in V \\ \|\mathbf{x}\|_2=1}} \|\mathbf{A}\mathbf{x}\|_2.$$

Definition 4 (Operator norm)

Suppose a vector norm $\|\cdot\|$ on $\mathbb{R}^m$ is given (e.g. the Euclidean norm).

Any $m \times n$ matrix $\mathbf{A}$ induces a linear operator from $\mathbb{R}^n$ to $\mathbb{R}^m$, and one defines the corresponding **operator norm** on the space $\mathbb{R}^{m \times n}$ of all $m \times n$ matrices as follows:

$$\begin{aligned}\|\mathbf{A}\| &= \sup\{\|\mathbf{A}\mathbf{x}\| : \mathbf{x} \in \mathbb{R}^n \text{ with } \|\mathbf{x}\| = 1\} \\ &= \sup\left\{\frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|} : \mathbf{x} \in \mathbb{R}^n \text{ with } \mathbf{x} \neq 0\right\}\end{aligned}$$

Examples of Operator Norm

- If the p -norm for vectors is used for both spaces $\mathbb{R}^n$ and $\mathbb{R}^m$, the corresponding induced operator norm is:

$$\|\mathbf{A}\|_p = \sup \left\{ \frac{\|\mathbf{Ax}\|_p}{\|\mathbf{x}\|_p} : \mathbf{x} \in \mathbb{R}^n \text{ with } \mathbf{x} \neq 0 \right\} \quad (1)$$

- For the case $p = 2$,

$$\|\mathbf{A}\|_2 = \sup \left\{ \frac{\|\mathbf{Ax}\|_2}{\|\mathbf{x}\|_2} : \mathbf{x} \in \mathbb{R}^n \text{ with } \mathbf{x} \neq 0 \right\} = s_1(\mathbf{A}),$$

where $s_1(\mathbf{A})$ is the largest singular value of $\mathbf{A}$ (*hint: use $\|\mathbf{Ax}\| = \langle \mathbf{Ax}, \mathbf{Ax} \rangle$*)

Notation

We will sometimes use the notation $\|\mathbf{A}\|_{\text{op}} := \|\mathbf{A}\|_2$ to denote the operator norm

Proposition 1

If $\mathbf{A}, \mathbf{B}$ are $n \times n$ Hermitian, then

$$|\lambda_i(\mathbf{A} + \mathbf{B}) - \lambda_i(\mathbf{A})| \leq \|\mathbf{B}\|_{op}.$$

proof:

We will prove

$$\lambda_i(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) + \|\mathbf{B}\|_{op}.$$

(we can prove the other inequality $\lambda_i(\mathbf{A}) - \|\mathbf{B}\|_{op} \leq \lambda_i(\mathbf{A} + \mathbf{B})$ using the same procedure)

By definition of operator norm,

$$\mathbf{v}^\top (\mathbf{A} + \mathbf{B}) \mathbf{v} = \mathbf{v}^\top \mathbf{A} \mathbf{v} + \mathbf{v}^\top \mathbf{B} \mathbf{v} \leq \mathbf{v}^\top \mathbf{A} \mathbf{v} + \|\mathbf{B}\|_{op}, \quad \forall \mathbf{v} \in \mathbb{R}^n \wedge \|\mathbf{v}\| = 1$$

Eigenvalue Stability Inequality (proof)

Given $V \subset \mathbb{R}^n$ where $\dim(V) = i$,

$$\min_{\mathbf{u} \in V, \|\mathbf{u}\|=1} \mathbf{u}^\top (\mathbf{A} + \mathbf{B}) \mathbf{u} \leq \mathbf{v}^\top (\mathbf{A} + \mathbf{B}) \mathbf{v} \leq \mathbf{v}^\top \mathbf{A} \mathbf{v} + \|\mathbf{B}\|_{\text{op}}, \forall \mathbf{v} \in V \wedge \|\mathbf{v}\| = 1,$$

then

$$\min_{\mathbf{u} \in V, \|\mathbf{u}\|=1} \mathbf{u}^\top (\mathbf{A} + \mathbf{B}) \mathbf{u} \leq \min_{\mathbf{v} \in V, \|\mathbf{v}\|=1} \mathbf{v}^\top \mathbf{A} \mathbf{v} + \|\mathbf{B}\|_{\text{op}}.$$

Taking the max over $\dim(V)$ on both sides, we obtain

$$\max_{\dim(V)=i} \min_{\mathbf{u} \in V, \|\mathbf{u}\|=1} \mathbf{u}^\top (\mathbf{A} + \mathbf{B}) \mathbf{u} \leq \max_{\dim(V)=i} \min_{\mathbf{v} \in V, \|\mathbf{v}\|=1} \mathbf{v}^\top \mathbf{A} \mathbf{v} + \|\mathbf{B}\|_{\text{op}}$$

which proves that $\lambda_i(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) + \|\mathbf{B}\|_{\text{op}}$

□

Frobenius Norm

Definition 5

The Frobenius norm of an $m \times n$ matrix $\mathbf{A}$ is defined as

$$\|\mathbf{A}\|_F := \left(\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2 \right)^{1/2} \quad (2)$$

Alternative definition

Equivalently, the Frobenius norm can be defined as

$$\|\mathbf{A}\|_F = \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle_F} = \sqrt{\text{tr} \mathbf{A}^\top \mathbf{A}} = \left(\sum_{i=1}^r s_i(\mathbf{A})^2 \right)^{1/2}, \quad (3)$$

where r is the rank of $\mathbf{A}$.

Norms and Singular values

Proposition

Let $\mathbf{s} = (s_1, \dots, s_r)$ be the vector of singular values of $\mathbf{A}$. Then

$$\|A\|_2 = \|\mathbf{s}\|_\infty, \quad \|A\|_F = \|\mathbf{s}\|_2.$$

Exercise

Show

$$\|A\|_2 \leq \|A\|_F \leq \sqrt{r} \|A\|_2.$$

Proposition

the extreme singular values control the distortion of the geometry of $\mathbb{R}^n$ under the action of the matrix $\mathbf{A}$:

$$s_n(\mathbf{A})\|\mathbf{x} - \mathbf{y}\|_2 \leq \|\mathbf{A}(\mathbf{x} - \mathbf{y})\|_2 \leq s_1(\mathbf{A})\|\mathbf{x} - \mathbf{y}\|_2$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Geometric Interpretation (Proof)

For $\mathbf{z} = \mathbf{x} - \mathbf{y} \neq 0$,

$$\|\mathbf{A}\mathbf{z}\|_2^2 = \mathbf{z}^\top \mathbf{A}^\top \mathbf{A} \mathbf{z} = \|\mathbf{z}\|_2^2 \frac{\mathbf{z}^\top}{\|\mathbf{z}\|_2} \mathbf{A}^\top \mathbf{A} \frac{\mathbf{z}}{\|\mathbf{z}\|_2}.$$

By slide 17,

$$\lambda_{\min}(\mathbf{A}^\top \mathbf{A}) \leq \frac{\mathbf{z}^\top}{\|\mathbf{z}\|_2} \mathbf{A}^\top \mathbf{A} \frac{\mathbf{z}}{\|\mathbf{z}\|_2} \leq \lambda_{\max}(\mathbf{A}^\top \mathbf{A}).$$

Taking square roots and using $s_n(\mathbf{A}) = \sqrt{\lambda_{\min}(\mathbf{A}^\top \mathbf{A})}$, $s_1(\mathbf{A}) = \sqrt{\lambda_{\max}(\mathbf{A}^\top \mathbf{A})}$ gives

$$s_n(\mathbf{A})\|\mathbf{z}\|_2 \leq \|\mathbf{A}\mathbf{z}\|_2 \leq s_1(\mathbf{A})\|\mathbf{z}\|_2.$$



Definition 6 (isometry)

If $\|\mathbf{A}\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for all $\mathbf{x} \in \mathbb{R}^n$, then $\mathbf{A}$ is called an isometry.

Equivalently, $s_n(\mathbf{A}) = s_1(\mathbf{A}) = 1$.

Interpretation

From a geometric point of view, it means that an isometry preserves distances.

Approximate Isometry

One can also define an **approximate** notion of isometry in the following sense:

Lemma 1 (Approximate isometry)

Let $\mathbf{A}$ be an $m \times n$ matrix such that

$$\|\mathbf{A}^\top \mathbf{A} - \mathbf{I}\|_2 \leq \delta,$$

for some $0 < \delta < 1$. Then

$$1 - \delta \leq s_{\min}(\mathbf{A}) \leq s_{\max}(\mathbf{A}) \leq 1 + \delta.$$

Remark: If $\mathbf{A}^\top \mathbf{A}$ is close to the identity, then all singular values of $\mathbf{A}$ are close to 1.

Approximate Isometry

proof:

Without loss of generality, we may assume that $\|\mathbf{x}\|_2 = 1$. Then, using the assumption, we get (by Cauchy-Schwarz)

$$\delta \geq \|\mathbf{A}^\top \mathbf{A} - \mathbf{I}\|_2 \geq |\langle (\mathbf{A}^\top \mathbf{A} - \mathbf{I})\mathbf{x}, \mathbf{x} \rangle| = ||\mathbf{Ax}\|_2^2 - 1|.$$

Applying the elementary inequality

$$\max(|z - 1|, |z - 1|^2) \leq |z^2 - 1|, \quad z \geq 0,$$

for $z = \|\mathbf{Ax}\|_2$, we conclude that

$$||\mathbf{Ax}\|_2 - 1| \leq \delta,$$

which implies the desired statement (using the min-max theorem for singular values)

□

Wigner Matrices and the Semicircle Law

Let $\mathbf{A}$ be an $n \times n$ **Wigner matrix** with eigenvalues $\lambda_1, \dots, \lambda_n$.

Goal

We will show that the empirical spectral distribution of $\tilde{\mathbf{A}} = \frac{1}{\sqrt{n}}\mathbf{A}$ converges to the **Wigner semicircle law**.

Why rescale?

To study the **asymptotic distribution of eigenvalues** as $n \rightarrow \infty$, we normalize the matrix:

$$\tilde{\mathbf{A}} = \frac{1}{\sqrt{n}} \mathbf{A}.$$

For a Wigner matrix $\mathbf{A} = (a_{ij})$, the entries are i.i.d. with

$$\mathbb{E}[a_{ij}] = 0, \quad \mathbb{E}[a_{ij}^2] = 1,$$

meaning that each entry of $\tilde{\mathbf{A}} = \frac{1}{\sqrt{n}} \mathbf{A}$ has variance $1/n$.

Hence, the expected squared norm of any row is

$$\mathbb{E}[\|\tilde{A}_{i\cdot}\|_2^2] = \sum_{j=1}^n \mathbb{E}[\tilde{a}_{ij}^2] = n \cdot \frac{1}{n} = 1.$$

Without this normalization, the eigenvalues would **blow up** with n .

Why eigenvalues blow up without rescaling (I)

Step 1: Expected value of $(\mathbf{Ax})_i^2$

For a unit vector $\mathbf{x}$ ($\|\mathbf{x}\|_2 = 1$), $(\mathbf{Ax})_i = \sum_{j=1}^n a_{ij}x_j$.

Then

$$\mathbb{E}[(\mathbf{Ax})_i^2] = \mathbb{E}\left[\left(\sum_{j=1}^n a_{ij}x_j\right)^2\right] = \sum_{j,k=1}^n x_jx_k \mathbb{E}[a_{ij}a_{ik}].$$

Because the entries are independent with $\mathbb{E}[a_{ij}] = 0$ and $\mathbb{E}[a_{ij}^2] = 1$,

$$\mathbb{E}[a_{ij}a_{ik}] = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$$

Hence $\mathbb{E}[(\mathbf{Ax})_i^2] = \sum_{j=1}^n x_j^2 = \|\mathbf{x}\|_2^2 = 1$. Each coordinate of $\mathbf{Ax}$ therefore has variance 1.

Why eigenvalues blow up without rescaling (II)

Step 2: Total expected squared norm

Summing over all i :

$$\mathbb{E}\|\mathbf{Ax}\|_2^2 = \sum_{i=1}^n \mathbb{E}[(\mathbf{Ax})_i^2] = n.$$

So, typically, $\|\mathbf{Ax}\|_2 \approx \sqrt{n}$.

Step 3: Consequence

Since

$$\lambda_{\max}(\mathbf{A}) = \max_{\|\mathbf{x}\|=1} \|\mathbf{Ax}\|_2,$$

it follows that $\lambda_{\max}(\mathbf{A}) \gtrsim \sqrt{n}$. Without normalization by $\frac{1}{\sqrt{n}}$, the eigenvalues **grow like** $\sqrt{n}$.

Wigner Matrix and Semi-circle Law

We will also require the following assumptions on the entries of the matrix $\mathbf{A}$:

Assumption 1

Assume that $\mathbf{A}$ is an $n \times n$ random matrix, whose entries a_{ij} are independent and such that for all $1 \leq i, j \leq n$,

- ▶ $\mathbb{E}[a_{ij}] = 0$,
- ▶ $\mathbb{E}[a_{ij}^2] = 1$,
- ▶ $\mathbb{E}[|a_{ij}|^k] < \infty$,
- ▶ The diagonal entries are identically distributed, and the off-diagonal entries are identically distributed

(these assumptions are for instance satisfied when $a_{ij} \stackrel{\text{i.i.d}}{\sim} \mathcal{N}(0, 1)$)

Wigner's Theorem (ESD convergence)

Theorem 3

Under Assumption 1, the eigenvalue distribution of Wigner matrices converges almost surely to Wigner's semi-circle as $n \rightarrow \infty$, in the sense that

$$\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{[a,b]}(\lambda_i) \xrightarrow[n \rightarrow \infty]{a.s.} \int_a^b d\mu_s(\lambda)$$

where $d\mu_s(\lambda) = \frac{1}{2\pi} \sqrt{4 - \lambda^2} d\lambda$ is the semicircle distribution on $[-2, 2]$.

(recall that a sequence of random variables $X_1, \dots, X_n$ defined on a sample space S converges almost surely to a random variable X , if $\Pr\left(\left\{s \in S : \lim_{n \rightarrow \infty} X_n(s) = X(s)\right\}\right) = 1$)

Wigner's Theorem – Illustration

For large n , the histogram of eigenvalues of $\frac{1}{\sqrt{n}}\mathbf{A}$ looks like this semicircle:

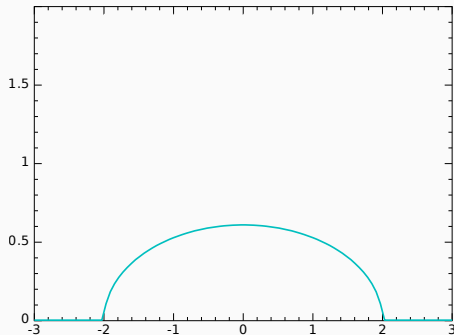


Figure 1: The distribution of eigenvalues λ of a symmetric random matrix with entries chosen from a standard normal distribution is illustrated above for a random 5000×5000 matrix

Recap: key takeaways

- ▶ **ESD**: equal mass at each eigenvalue.
- ▶ **Min–Max**: variational characterization of eigen/singular values.
- ▶ **Norms**: $\|A\|_{\text{op}} = s_1(A)$, $\|A\|_F^2 = \sum s_i^2$.
- ▶ **Wigner** $\Rightarrow$ **Semicircle**; **Wishart** $\Rightarrow$ **Marchenko–Pastur** (next).

References

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